

LECTURE 2

Topics

Financial intermediaries, Security issuance process, Mechanics of trading, Regulation of securities markets

- The allocation of capital from capital suppliers to capital demanders is facilitated by financial intermediaries
- The main financial intermediaries are:
 - Commercial banks
 - Credit unions
 - Investment banks
 - Investment companies
 - Insurance companies
- The main suppliers of capital are consumers that want to save/invest a portion of their income
- The main demanders of capital are corporations and governments

- Commercial banks and credit unions raise capital through term deposits and allocate it typically to corporations in the form of loans
- Recall that consumers that want to save a portion of their income do so by acquiring term deposits
- Commercial banks are for-profit institutions that take actions to maximize shareholder value
- Credit unions are non-profit organizations owned by their customers ("members") and take actions to maximize benefits for their members
- Non-profit status exempts credit unions from federal taxes and these tax savings are passed on to members in the form of lower fees, higher interest rates on term deposits, and lower interest rates on loans to members
- Commercial banks offer a one-stop shopping experience for consumers who also want to invest

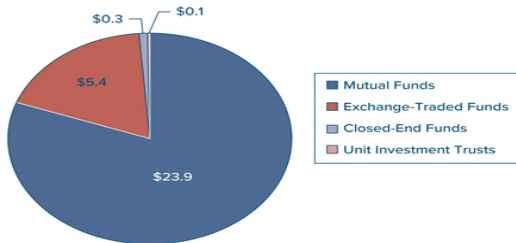
- Investment banks facilitate corporations to raise capital by issuing securities such as common stock and corporate bonds
- Corporations that want to issue securities need an investment bank to underwrite that issuance
- The investment bank advises the corporation on pricing and sells the securities issued to the public
- The newly issued securities are sold to the public on the so-called *primary market*
- Investors can then trade these securities among themselves on the *secondary market*
- The distinction between primary and secondary markets is about the nature of the securities traded: newly-created vs pre-owned securities
- Selling of securities on primary market is heavily regulated

- Glass-Steagal Act of 1933-1999 prevented commercial banks from engaging in investment banking activities
- Commercial banks were supposed to be "boring"
- Unlike investment banks, commercial banks are part of the Federal Reserve System and have access to additional sources of liquidity
- This could be valuable in a financial crisis when liquidity tends to dry up

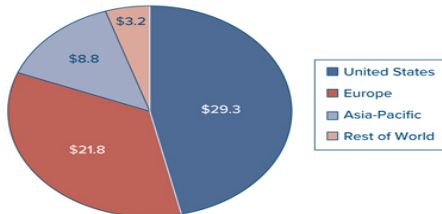
- Another class of intermediaries is investment companies
- An investment company is a corporation that invests the pooled capital of investors in financial securities
- We distinguish between public vs private investment funds
- *Public investment funds* are heavily regulated and opened to all investors:
 - Open-ended funds ("mutual funds")
 - Closed-ended funds
 - Exchange-traded funds
 - Unit investment trusts
- *Private investment funds* are less regulated and open only to wealthy ("accredited") investors:
 - Hedge funds
 - Private equity funds
 - Venture capital funds

Public investment funds in the US

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Investment Company Assets Under Management, United States (\$ Trillion)



Assets Under Management, Open End Funds (Assets in \$ Trillions)



- Open-ended funds are portfolios of financial assets financed by issuing shares that are then sold to investors
- Shares in a mutual fund are *issued* and *redeemed* as needed at the fund's *net asset value* (NAV)
- When an investor wants to cash out, it simply sells its shares back to the mutual fund at NAV
- A mutual fund's NAV is calculated at the end of each trading day:

NAV Formula

$$\text{NAV} = \frac{\text{Value of the fund's asset portfolio net of liabilities}}{\text{Number of shares outstanding}}$$

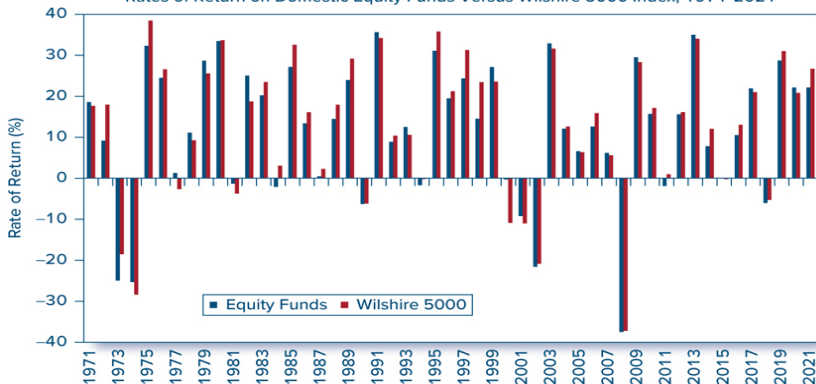
- Each mutual fund has a board of directors that hires a management company to manage the fund's assets
- The management company charges the fund a fee that ranges in 0.2% - 1.25% of assets under management
- The mutual fund passes on this management fee to the investors

- A typical mutual fund must state its investment policy in the fund's prospectus to investors
- Money market funds: Invest in money market securities such as commercial paper, repurchase agreements, or CDs
- Equity funds: Invest primarily in stock
- Sector funds: Concentrate on a particular industry or country
- Bond funds: Specialize in the fixed-income sector
- International funds: Global, international, regional, and emerging market
- Balanced funds: Designed to be candidates for an individual's entire investment portfolio
 - Hold both equities and fixed-income securities in relatively stable proportions
 - Many are *funds of funds*

- Mutual funds can have high levels of risk, depending on their investment strategy
- They often engage in market timing strategies
- The SEC generally limits the extent to which mutual funds can use leverage either through loans or through the use of derivatives
- Mutual funds are subject to quarterly reporting requirements on their holdings
- A mutual's fund performance is usually measured by comparing the fund's annual rate return net of fees against those of a relevant market index benchmark
- For equity funds the relevant benchmark is the Wilshire 5000
- Wilshire 5000 average return was 12.49%, which was 0.96% greater than the average mutual fund from 1971 to 2020

Equity funds vs Wilshire 5000

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Rates of Return on Domestic Equity Funds Versus Wilshire 5000 Index, 1971-2021



- Mutual funds are not very tax-efficient for their investors
- "Pass-through status" under the U.S. tax code
 - Taxes are paid only by the investor, not by the fund itself
 - Disadvantage is that fund investors do not control the timing of the sales of securities from the portfolio, reducing their ability to engage in tax management
- High portfolio turnover rate (a measure of how frequent fund managers buy and sell securities) can be particularly "tax inefficient"

- Mutual funds are distributed through several channels
- Directly by the fund underwriter (direct-marketed funds)
 - Sold through the mail, various offices of the fund, over the phone, or over the Internet
- Indirectly through brokers acting on behalf of the underwriter (sales-force distributed)
 - Broker or financial advisers receive a commission for selling shares
 - This distribution channel raises potential conflict of interest for brokers that provide investment advice to customers
- Financial supermarkets
 - Sell shares in funds of many complexes
 - Broker splits management fees with the mutual fund company

- Investing in mutual funds comes with fees
- Mutual funds must disclose these fees in their prospectuses to the investors
- Fee structure
 - 1 Operating expenses: Admin expenses and advisory fees paid to investment management company
 - 2 Front-end load: A fee paid upfront when buying the fund
 - 3 Back-end load: A fee paid when selling the fund
 - 4 12b-1 fees: A fee that covers marketing and distribution of the fund (capped by law at 0.75%)
- A fund's expense ratio = operating expenses + 12b-1 fees, and are deducted annually from the fund's assets
- Many funds offer *classes* of shares that represent ownership in the same portfolio of assets, but with different mixes of fees

- Share classes and their fees at Dreyfus High-Yield Fund in 2021
- Notice the fee trade-offs in Class A and Class C shares
- Class I shares are sold only to institutional investors

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	Class A	Class C	Class I
Front-end load	0–4.5% ^a	0	0
Back-end load	0	0–1% ^b	0%
12b-1 fees	0.25%	1.0% ^c	0%
Expense ratio	0.7%	0.7%	0.7%

^aDepending on size of investment. Starts at 4.5% for investments less than \$50,000 and tapers to zero for investments more than \$1 million.

^bDepending on years until holdings are sold. Exit fee is 1% for shares redeemed within one year of purchase.

^cIncluding annual service fee.

- Fees make a serious dent in a fund's rate of returns
- A fund's rate of return before fees is calculated as follows:

Fund's annual return before fees

$$r_1 = \frac{\text{NAV}_1 + d_1 - \text{NAV}_0}{\text{NAV}_0}$$

- d_1 is the income generated by the fund (e.g., dividends and capital gains) over the holding period
- A fund's rate of return after expense fees, f , is:

$$r_1^* = r_1 - f$$

- The front load affects the initial invested capital, I_0

$$I_0^* = I_0 \times (1 - L_{\text{Front}})$$

- The back load affects the terminal invested capital, I_T

$$I_T^* = I_T \times (1 - L_{\text{Back}})$$

- An equity fund sells 3 share classes with different fee structure:
 - Class A: Front load of 3%, 12b-1 fees of 0.25% per year, and no back-end load fees
 - Class B: Front load of 1%, 12b-1 fees of 0.75% per year, and back-end load fees of 0.5%
 - Class C: 12b-1 fees of 0.5% per year, and back-end load that starts at 5% in the first year and drops by 1% after each holding year
- The fund's operating expenses are 0.7%
- If the fund has an expected rate of return of 10% per year before fees, what would be the value of \$10,000 in class A, B or C, if the fund shares were to be sold after
 - 1 year
 - 4 years
 - 10 years

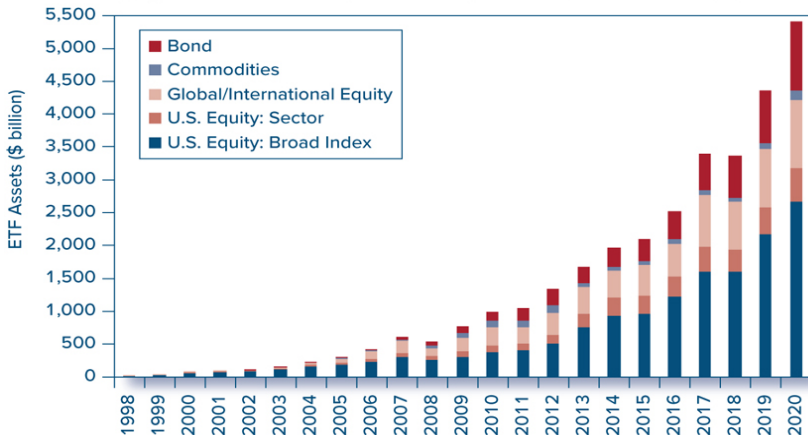
- A special class of mutual funds are Money Market (MM) funds
- Money market funds are mutual funds that invest in money market instruments
- There are two types of MM funds:
 - *Government funds* hold short-term U.S. Treasury or agency securities
 - *Prime funds* also hold other money market instruments such commercial paper
- MM funds generally aim to maintain an NAV of \$1 per share and all excess earnings that MM fund generates is returned to investors as dividend payments
- MM funds are often viewed by investors as substitutes for bank savings accounts
- However, unlike guaranteed bank accounts, MM funds are risky and may lose money ("breaking the buck")

- Closed-end funds are funds that issue a *fixed* number of shares to finance their portfolio of assets
- These shares trade on exchanges
- Thus, when an investor wants to cash out it simply sells its shares on the exchange
- The price of a share on the exchange may differ from the fund's NAV
- These differences can persist for long periods of time, and they are a bit of a puzzle for economists
- For example if fund shares trade at a discount relative to fund's NAV, the fund could simply sell all its assets at NAV and create immediate wealth for its shareholders

- Exchange-traded funds (ETFs) combine the defining features of open-ended and closed-ended mutual funds
- Like mutual funds, an ETF issues/redeems shares as needed to finance its portfolio of assets
- Like closed-end funds, ETFs shares trade on exchanges
- Unlike mutual funds, ETFs do not issues/redeem shares continuously, but rather in *blocks of shares* ("creation units")
- ETFs only issue and redeem shares to a certain class of specialized large investors called *authorized participants* or APs
- When ETF wants to issue new shares
 - It asks its APs to buy a "slice" of ETF's portfolio of assets in the market (at NAV) and sell it to the ETF in exchange for ETF shares
 - The APs then sell the ETF shares in the market, for a profit

ETF Assets in the US

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- ETFs are of two types: actively or passively managed
- *Actively-managed* ETFs hire a management firm to manage the ETF's portfolio of assets
- *Passively-managed* ETFs track certain market indexes
- ETFs are popular because they provide a tradable alternative for non-tradable indexes
 - *Market* ETFs track market indexes such as S&P 500
 - *Sector or industry* ETFs track sector or industry indexes such as utilities, pharmaceuticals, etc.
 - *Currency* ETFs track currencies such as the US dollar
 - *Commodity* ETFs tracks commodities such as oil, gold etc.
 - *Bitcoin* ETFs buy and hold Bitcoins for their investors
 - *Leveraged* ETFs provide 2X or 3X exposure to an index using derivatives

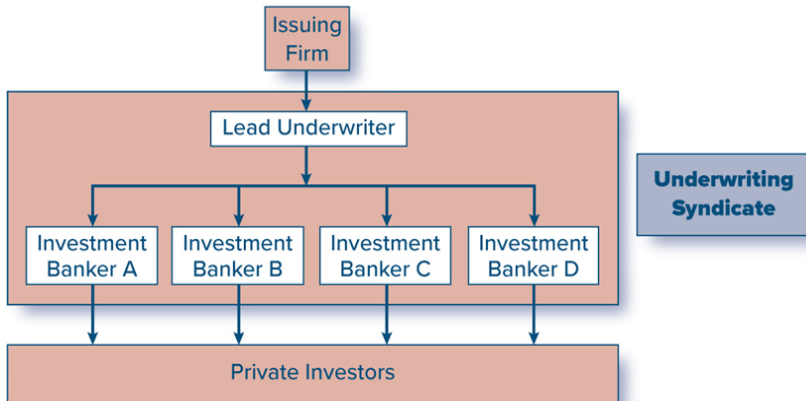
- Hedge funds are funds structured as private partnerships and open to accredited investors only
- They can be either open-ended or closed-ended funds
- They pursue *investment strategies* that are outside the reach of most mutual funds due to regulation constraints
- **Venture capital funds** invest exclusively in young private firms (e.g., start-ups) with high growth potential
- **Private equity funds** take equity stakes in private firms that are already established (mature) for the purpose of taking them public
- All these funds are viewed by institutional investors as an *alternative* asset class

- *Private firms* are owned by a relatively *small* number of shareholders
- They raise capital by selling securities directly to institutional investor in *private placements*
- Originally, number of shareholders was capped at 499, but the Jumpstart Our Business Startups (JOBS) Act of 2012 increased the cap to 2000
- The JOBS Act also made it easier for these firms to raise capital through *crowdfunding*
- If number of shareholders *exceeds* 2000, the private firm has to *register the stock with the SEC* and be subject to *reporting requirements*
- In general, private firms are not subject to the stringent reporting requirements of public firms
- Securities issued in private placements do not trade on exchanges, and thus can be illiquid

- Public firms can issue new securities such as stock and bonds in either
 - *Initial Public Offering (IPO)*: The first time the firm issues securities to the public
 - *Secondary Public Offering*: A follow up issuance of securities
- A Secondary public offering for a firm's stocks is also known as *seasoned equity offering* (SEO)
- Public offerings are *underwritten* by investment banks that provide valuation advice for the issuer and help market the securities
- If the offering is large, the underwriters may form a *syndicate* to reduce exposure to the issuer

- The issuance process

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- Underwriters organize *road shows* to market the offering but also get a sense of the demand for the security ("bookbuilding")
- Road shows are opportunities to learn about the price at which securities should be sold to the public
- In an IPO there is great uncertainty about the price of the security offered, and much that rides on these prices
 - Issuer needs to raise the much needed capital
 - Underwriter keeps a portion of offering and thus is exposed to price changes
 - Large investors also have significant exposures
- Underwriter's reputation is at stake: **underpricing hurts the issuer**, **overpricing hurts the investors**
- As a result of these issues, and others, underwriters tend to *underprice* the offering
- The underpricing is a cost that adds to the underwriting fee (7%)

- Firms have two alternatives to going public via an IPO
 - Direct listing
 - Special Purpose Acquisition Company (SPAC)
- In a *direct listing*, existing shareholders of the private issuer sell their shares directly to the public
- Unlike an IPO, in a direct listing the firm doesn't issue additional shares and doesn't need an underwriter (so it saves on the cost of underwriting = fee + underpricing)
- Direct listing seems more suited for firms that do not need to raise capital right away
- The sponsor of a *SPAC* raises funds in its own IPO and goes public, but with no underlying commercial operations
- Searches for an acquisition target, merging it into the publicly traded SPAC ("Blank-Check Firms")
- After 2 years, if there is no suitable acquisition, the money must be returned to investors

- Firms that are already public can raise additional capital via:
 - SEOs
 - Shelf Registration
- SEOs are underwritten offering of additional securities
- Unlike IPOs, SEOs have a better sense of the price of the offering because the stock is already trading
- All securities sold to the public (IPOs or SEOs) must be registered with the SEC
- *Shelf registration* is an alternative way to raise capital for a public firm without incurring underwriting costs
- Allows firms to register securities and gradually sell them to the public for 3 years following the initial registration
- Shares can be sold on short notice and in small amounts without incurring underwriting costs

- Financial assets trade on organized markets such as exchanges and OTC markets
- Most of these markets are intermediated by *brokers and dealers*
- Dealers hold an *inventory* of certain financial assets and stand ready to make a market in those assets
- Dealers often quote bid and ask prices for their assets
 - **Bid price** is the price at which the dealer is willing to buy
 - **Ask price** is the price at which the dealer is willing to sell
- Because carrying an inventory comes with risk of price drops, dealers always set ask prices above bid prices
- The difference between the two is called the *bid-ask spread* and is a measure of dealer compensation

- Brokers help match buyers and sellers in exchange for a commission fee
- Unlike dealers, brokers usually do not carry inventory of assets
- One way brokers help match buyers and sellers is by facilitating access to investors to electronic exchanges
- On an electronic exchange, buyers and sellers are matched electronically
- Brokers can be of two types:
 - Discount: Buy/sell on behalf of customers, hold customers assets, provide margin loans
 - Full-service: Same as Discount + provide advice to customers on trades
- Full-service brokers are subject to additional reporting meant to help investors understand how the broker benefits from the transactions on which it provides advice (Regulation Best Interest)

- Trading an asset always involves costs
- Explicit costs:
 - Brokerage commissions for executing buy or sell orders
 - Interest on margin loans and short sales
 - Pay for advice about investment alternatives
- Implicit costs
 - Dealer's bid-ask spread
 - Price impact from the brokers selling their entire order flow from all their customers to financial intermediaries
 - Price concession associated with trading more than the quantities associated with the posted bid or ask price
 - A portion of the fees charged by mutual fund products could go to the brokers that advised customers on purchasing the fund in the first place

- An investor is buying an asset on margin when it borrows part of the purchase price from a broker
 - **Initial Margin**: Investor's contribution to the purchase of the asset
 - **Margin loan**: the amount borrowed from the broker
 - **Equity**: Current value of the asset net of the margin loan
- At the outset, the initial margin equals the account equity
- If the equity turns negative, investor may default on the loan
- To minimize the risk of losing money on the loan, the broker requires that the investor maintains equity above a certain level, known as **maintenance margin**
- When the equity falls below the maintenance margin, the broker will ask the investor to add more cash to its margin account ("margin call")

- To buy on margin an investor has to set up a margin account with the broker
- Margin requirements are regulated by FRB and FINRA
 - FRB: Initial margin has to be larger than 50% (investors can borrow at most 50% of the value of the purchase)
 - FINRA: Maintenance margin has to be at least 25%
- Brokers will set their own maintenance margin (called house maintenance margin) that is larger than the FINRA's minimum
- Margin account equity equals

Account equity

$$\text{Equity} = \frac{\text{Value of position} - \text{Margin loan}}{\text{Value of position}}$$

- Suppose a stock is currently worth \$100 per share
- An investor wants to buy 200 shares of the stock on margin with an initial margin (IM) of 50%
- The broker has a house maintenance margin (MM) of 30%
- Purchase value of the stock is

$$200 \times \$100 = \$20,000$$

- Investor's Initial Margin (IM) is

$$\$20,000 \times 50\% = \$10,000$$

- Margin loan principal equals:

$$\$20,000 - \$10,000 = \$10,000$$

- Margin account equity is

$$\text{Equity} = \frac{\$20,000 - \$10,000}{\$20,000} = 50\%$$

- Suppose now that the stock price goes down to \$70
- Margin account equity is

$$\text{Equity} = \frac{200 \times \$70 - \$10,000}{200 \times \$70} = \frac{\$4,000}{\$14,000} = 28.57\%$$

- Is Equity > MM?: No!

$$28.57\% < 30\%$$

- Investor gets margin call
- Must add cash to margin account in the amount of:

$$(30\% - 28.57\%) \times \$14,000 = \$200.2$$

- The margin account is also going to be used to make the interest payment on the margin loan
- The interest rate on the margin loan is called *margin interest*
- To extend a margin loan to a customer, the broker usually borrows the funds from a bank
- The broker's loan to the bank is called *broker's call loan*
- This loan is short-term and callable (the bank can call the loan back at any moment)
- The broker may increase its MM suddenly to meet the bank's request
- The margin interest on a margin loan equals the interest on the broker's call loan plus a fee

- **Selling securities short** ("Short-selling") allows investors to profit from a decline in a security's price
- In a short sale, the investor borrows securities from a broker and sells them right away
- The proceeds from the sale must be kept in a margin account with the broker
- To close the position ("cover the short"), the investor purchases the securities in the market and returns them to the broker
- The investor would have to post margin for the borrowed securities
- In short-selling, the proceeds from selling the securities short net of margin triggers a *margin loan* on which the borrower has to pay margin interest
- In addition to returning the securities, the investor also has to compensate the broker for any dividends or coupon payments that the borrowed securities paid while on loan

- Suppose a stock is currently worth \$100 per share
- An investor believes the stock will go down in the near future and wants to sell short 1000 shares of the stock
- The broker requires an initial margin (IM) of 50% on short sale positions and a house maintenance margin (MM) of 30%
- To establish the short position, the investor has to post collateral:

$$IM = 50\% \times 1000 \times \$100 = \$50,000$$

- Proceeds from selling the borrowed stock:

$$1000 \times \$100 = \$100,000$$

- Margin account equity becomes

$$\text{Equity} = \frac{\$100,000 + \$50,000 - 1000 \times \$100}{1000 \times \$100} = \frac{\$50,000}{\$100,000} = 50\%$$

- Suppose now that the stock price goes down to \$70
- Investor closes out the position by purchasing 1000 shares of the stock on the market
- Cost of closing out the position to the investor is:

$$1000 \times \$70 = \$70,000$$

- The investor returns the shares to the broker
- The margin account is going to show:
 - A cash profit of:

$$\$100,000 - \$70,000 = \$30,000$$

- The initial margin of:

$$IM = \$50,000$$

- Suppose now that the stock price goes up to \$120
- Margin account equity is

$$\text{Equity} = \frac{\$100,000 + \$50,000 - 1000 \times \$120}{1000 \times \$120} = \frac{\$30,000}{\$120,000} = 25\%$$

- Is Equity > MM?: No!

$$25\% < 30\%$$

- Investor gets margin call
- Must add cash to margin account in the amount of:

$$(\$30,000 - \$25,000) = \$5,000$$

- The main laws protecting investors in the US are:
 - Securities Act of 1933
 - Securities Exchange Act of 1934
 - Securities Investor Protection Act of 1970
 - *Blue sky* laws
- US Securities and Exchange Commission (SEC) regulates the issuance and trading of all financial assets that are deemed to be *securities*
- Commodity and Futures Trading Commission (CFTC) regulates the trading of all financial assets deemed to be commodities or futures
- The main self-regulating body for brokers and dealers is the Financial Industry Regulatory Authority (FINRA)

- Sarbanes-Oxley Act (SOA) came out as a result of the 2000-2002 accounting scandals:
 - Allocations of shares in IPOs
 - Tainted securities research and recommendations
 - Misleading financial statements and accounting practices
- The key provisions of SOA are:
 - Public Company Accounting Oversight Board
 - Independent financial experts to serve on audit committees of a firm's board of directors
 - CEOs and CFOs personally certify firms' financial reports
 - Auditors may no longer provide several other services to clients
 - Boards must have independent directors

- The Regulations prohibit trading on inside information
- SEC requires officers, directors, and major stockholders to report all transactions in their firm's stock
- Insiders do exploit their knowledge
 - Well-publicized convictions of principals in insider trading schemes
 - Considerable evidence of "leakage"
 - Documented abnormal returns on trades by insiders